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Dose-dependent effects of micro-aeration on methane production and lignocellulose degradation during continuous anaerobic digestion of food waste

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ABSTRACT

While micro-aeration is a promising strategy to enhance hydrolysis and acidogenesis in anaerobic digestion (AD), its dose-dependent effects on lignocellulose degradation and methane production under different total solids (TS) conditions remain unclear. This study investigated continuous AD of food waste (FW) at TS levels of 6% and 10% under oxygen loading rates of 0, 2.625, 5.25, and 10.50 $mLO_2/L_R/d$. At TS = 6%, 2.625 $mLO_2/L_R/d$ modestly increased methane yield from 527.2 to 561.9 mL/gVS_{added} and increased cellulose removal to 60.16%. At 5.25 $mLO_2/L_R/d$, biogas production increased further, whereas methane content decreased, indicating enhanced upstream substrate conversion rather than proportionally enhanced methanogenesis. At TS = 10%, improved methane production was observed only at 2.625 $mLO_2/L_R/d$, while higher oxygen loading impaired process performance. Excessive aeration caused volatile fatty acid (VFA) accumulation, increased oxidation reduction potential (ORP), and reduced lignocellulose degradation. Microbial community analysis showed oxygen-dependent shifts in lignocellulose-associated bacteria and methanogens, while PICRUSt2-based prediction suggested changes in hydrolysis-related functions, energy metabolism, and oxidative-stress-response potential. Overall, the effective oxygen-loading window narrowed with increasing TS, and low oxygen input modestly improved methane production, whereas moderate oxygen loading mainly enhanced upstream substrate conversion.

1. Introduction

FW management has become a critical issue in sustainable waste treatment, resource recovery, and greenhouse gas mitigation. In 2022, approximately 1.05 billion tonnes of food were wasted globally, contributing an estimated 8–10% of global greenhouse gas emissions [1,2]. Anaerobic digestion is widely recognized as an effective technology for FW treatment because it enables organic stabilization, methane recovery, and nutrient recycling [3,4]. However, FW contains lignocellulosic fractions such as vegetable stalks and fruit peels, in which cellulose, hemicellulose, and lignin are embedded in a recalcitrant matrix that limits enzymatic accessibility and hydrolysis efficiency [5,6]. As hydrolysis is generally regarded as the rate-limiting step in FW AD, improving lignocellulose solubilization while maintaining stable methanogenesis is essential for enhancing the overall conversion of FW into methane.

Several studies have explored ways to improve the performance of AD, including substrate pretreatments, optimizing process parameters and introducing additives. However, the pretreatment or supplementary additives usually required extra energy and chemicals, thereby increasing operational costs and process complexity and potentially causing environmental concerns. Recent studies have shown that micro-aeration, which involves injecting small amounts of air or oxygen into an anaerobic system, has attracted increasing attention as a cost-effective strategy for improving digestion performance by accelerating hydrolysis and acidogenesis and providing more available substrates for

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methanogenesis [7–11]. Recent studies have also highlighted the potential of micro-aeration in continuous FW AD, while indicating that solids content, viscosity, and reactor stability may influence its effectiveness [12]. Moderate oxygen exposure can stimulate hydrolytic and facultative microorganisms to secrete higher levels of extracellular enzymes (e.g., cellulases and xylanases). This enzymatic enhancement accelerates the disruption and degradation of complex lignocellulose, thereby providing more readily available substrates for methanogenesis. In FW AD, micro-aeration has been shown to affect hydrolysis, VFA dynamics, methane production, and microbial community structure [13, 14]· Moreover, appropriate oxygen loading can enhance methane production and lignocellulose degradation, whereas excessive aeration may impair methanogenesis despite promoting organic matter removal [15, 16]. However, the interaction between TS content and oxygen loading in continuous food-waste anaerobic digestion remains insufficiently understood. These findings indicate that the effectiveness of micro-aeration depends strongly on oxygen dosage, substrate properties, and reactor operating conditions.

The effects of micro-aeration are inherently dose dependent. Appropriate oxygen input can be rapidly consumed by facultative microorganisms, creating localized redox gradients that favor hydrolysis and acidogenesis while preserving anaerobic niches for methanogenesis. In contrast, excessive oxygen may inhibit strictly anaerobic methanogens, promote oxidative mineralization of organic carbon, and lead to VFA accumulation or methane-yield reduction [17,18]. Therefore, the central challenge is to define an appropriate oxygen-loading window that enhances hydrolysis without disrupting downstream methanogenesis. This challenge is particularly relevant in continuous FW AD, where accelerated hydrolysis may increase soluble organic matter and VFA formation, whereas methanogenesis may respond more slowly, leading to decoupling between hydrolysis and methanogenesis kinetics. Although previous studies have shown that micro-aeration can regulate substrate hydrolysis, VFA production, methane generation, and microbial communities, its effects depend strongly on oxygen input and reactor operating conditions [19,20]. Whether the favorable oxygen-input range becomes narrower as reactor loading increases, and how this is associated with the balance between upstream substrate conversion and downstream methanogenesis, remains insufficiently characterized. Because TS and OLR covary in many practical high-solids digestion systems, understanding their combined operational context is important for defining oxygen-input strategies without over-attributing the observed response to either factor alone. Therefore, a better understanding of the TS-dependent response to micro-aeration is needed to identify an appropriate oxygen-loading window while avoiding adverse effects on methanogenesis.

In this study, continuous AD of FW was conducted under two TS levels (6% and 10%) with different micro-aeration loading rates. The objectives were to (i) characterize the responses of continuous FW AD to increasing oxygen loading at 6% and 10% TS levels; (ii) assess their effects on methane production, process stability, VFA dynamics, SCOD, and lignocellulose degradation; and (iii) develop a proposed interpretation of the process responses through microbial community and predicted functional analyses. Attention was given to the balance between enhanced upstream substrate conversion and maintenance of downstream methanogenesis under different oxygen-loading conditions. This study characterizes the responses under two TS conditions at a similar nominal organic loading rate (OLR) and provides a basis for selecting oxygen inputs to enhance lignocellulose conversion and methane recovery while maintaining stable methanogenesis.

2. Materials and methods

2.1. Substrate and inoculum

To ensure consistency in the experiment, the FW substrates were prepared in the ratio of cabbage: celery: tofu: meat: rice= 2:2:1:1:2. Before further use, FW was chopped. The main characteristics of FW used were as follows: TS of 13.33% ± 0.84, volatile solids (VS) of 95.22% ± 1.21 (based on TS), soluble chemical oxygen demand (SCOD) of 30550 ± 125.00 mg/L, pH of 7.63 ± 0.12. Sludge from AD of domestic waste served as inoculum and was taken from the Beijing Dongcun comprehensive treatment plant. The main characteristics are as follows: TS of 7.51% ± 0.96, VS of 53.50% ± 0.84 (based on TS), SCOD of 8715 ± 54.37 mg/L, pH of 8.46 ± 0.10.

2.2. Continuous methane production from food waste via micro-aerobic-enhanced anaerobic digestion

2.2.1. Reactor configuration and operating parameters

Continuous AD experiments were conducted in double-jacketed glass reactors with a total volume of 5 L and a working volume of 4 L, maintained at 37 ± 1 °C. The daily feeding and withdrawal volume were 140 and 84 mL for the TS = 6% and TS = 10% systems, corresponding to a hydraulic retention time (HRT) of 28.6 d and 47.6d, respectively. The reactors were fed once daily, with an OLR of approximately 2.00 g VS/L/d for both the TS = 6% and 10% systems. Each reactor was mechanically stirred at 20 rpm and equipped with a top feed port and biogas outlet and a bottom effluent outlet. After achieving stable anaerobic operation, the reactors were maintained for an additional 5-day stabilization period for this experiment, and operational stability was defined as relatively stable daily gas production for 3 consecutive days.

2.2.2. Selection of TS levels and oxygen-loading rates

Based on preliminary batch experiments, TS levels of 6% and 10% were selected and designated as R1 and R2, respectively. These levels represent different solids-loading conditions. These two TS levels were used to represent different solids-content conditions while maintaining a similar nominal OLR of approximately 2.00 g VS/L/d. The 6% TS system was expected to have greater fluidity and more favorable mass transfer, whereas the 10% TS system was expected to have lower fluidity and greater sensitivity to oxygen-input fluctuations. Four consecutive oxygen-loading phases were established at 0, 2.625, 5.25, and 10.5 $mLO_2/L_R/d$. The oxygen-loading rates were selected based on previous studies and preliminary experiments to cover low, intermediate, and relatively high oxygen inputs. The anaerobic phase was designated CK, while the three micro-aeration phases were designated T1, T2, and T3, respectively. For each TS condition, two parallel reactors were operated continuously for 60 d, with each oxygen-loading phase maintained for 15 d. The corresponding operational phases were designated R1_1–R1_4 and R2_1–R2_4 for the TS = 6% and 10% systems, respectively.

2.2.3. Aeration delivery

Air was supplied directly into the digestate using a peristaltic pump (BT100–2J/YZ) at 0.5 mL air/min. Oxygen loading = air flow rate × aeration duration × 0.21/working volume. Aeration was conducted four times per day at 6 h intervals, with each aeration event lasting 25, 50, and 100 min for T1, T2, and T3, respectively, corresponding to total daily aeration durations of 100, 200, and 400 min/d. The produced biogas was collected in 10-L aluminum foil gas sampling bags (45 cm × 35 cm; Changde BKM Biological Technology Co., Ltd.).

2.2.4. Sampling schedule

pH and ORP were monitored daily. Liquid samples were collected every 3 d for SCOD, TCOD, and VFA analyses. Microbial community samples were collected once during the late, operationally stable period at the end of each 15-d operational phase.

2.3. Analysis methods

TS, VS, ammonia nitrogen (NH_4^+-N) and SCOD were measured by standard methods [21]. pH and ORP were measured with a portable pH meter (SX620, Shanghai) and portable ORP meter (SX630, Shanghai).

The volume of biogas production was tracked using the water displacement method. The methane content and VFA were determined by a gas chromatograph (GC-2014C, Shimadzu, Japan).

2.4. Microbial analysis

Microbial community samples were collected once during the late, operationally stable period at the end of each 15-d operational phase. 2 mL of digestion broth was centrifuged (15,000 rpm, 10 min) and the sediment used for microbial community analysis. Genomic DNA was extracted with the PowerSoil DNA Isolation Kit and verified by 1% agarose gel electrophoresis. The bacterial 16S rRNA V3–V4 and archaeal V4–V5 regions were PCR-amplified using 515FmodF and 806RmodR primers, respectively, quantified with the QuantiFluor™-ST system, and sequenced on Illumina MiSeq (Majorbio) after TruSeq™ library construction. Functional profiles and the relative abundance of predicted enzyme-related genes were inferred using PICRUSt2. (http://huttenhower.sph.harvard.edu/galaxy).

2.5. Statistical analysis

Pearson correlation analysis was performed to evaluate the relationships among physicochemical parameters, AD performance, lignocellulose degradation, and process efficiencies. Correlation coefficients (Pearson's r) range from −1 to 1, with positive and negative values indicating positive and negative correlations, respectively. Statistical significance was determined at $P < 0.05$. Mantel tests based on Bray–Curtis dissimilarity were used to evaluate the associations between bacterial or archaeal community composition and environmental and process variables.

3. Results and discussion

3.1. The effect of micro-aeration conditions on methane production of food waste

Fig. 1 investigates the effects of varying oxygen loads on gas production in continuous AD reactors treating FW at TS levels of 6% and 10%. At TS = 6%, both biogas and methane yields increased at oxygen loads of 2.625 and 5.25 $mLO_2/L_R/d$. The methane yields reached 561.9 and 560.2 mL/gVS_{added}, respectively, compared with 527.2 mL/gVS_{added} under anaerobic conditions. However, further increasing the oxygen load to 10.50 $mLO_2/L_R/d$ resulted in a marked decrease in methane yield to 382.3 mL/gVS_{added}, corresponding to a 27.49% decrease relative to the anaerobic condition.

At TS= 10%, an improvement in methane production was observed only at an oxygen load of 2.625 $mLO_2/L_R/d$, where the methane yield increased to 648.7 mL/gVS_{added}, compared with 530.3 mL/gVS_{added} under anaerobic conditions. In contrast, methane yields decreased to 524.9 and 391.0 mL/gVS_{added} at oxygen loads of 5.25 and 10.50 $mLO_2/$

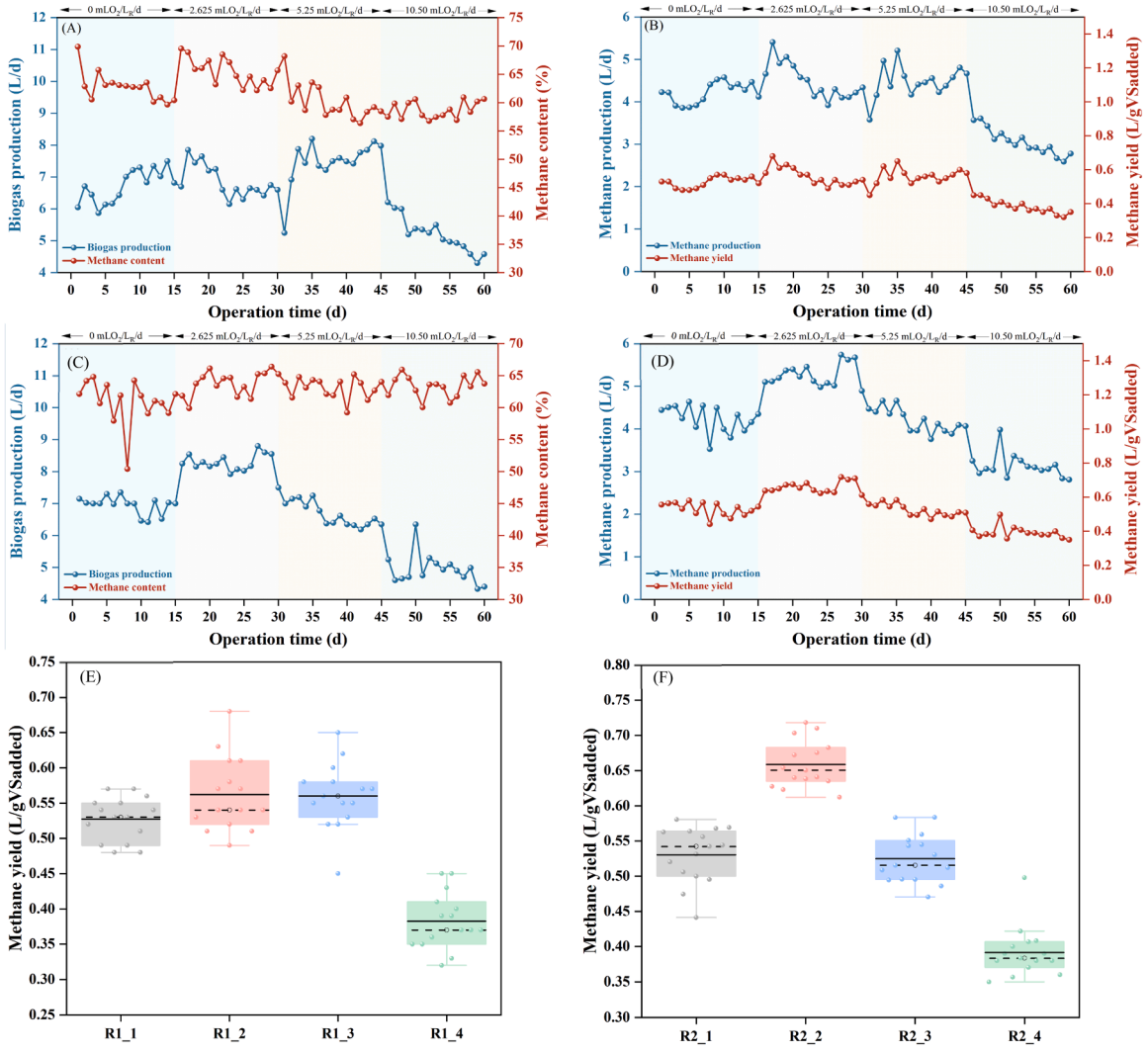


Fig. 1. Characteristics of biogas and methane production, methane content, and methane yield at TS levels of 6% and 10%.

L_R/d, respectively. These results indicate that low micro-aeration promoted methane production, whereas excessive oxygen loading inhibited methanogenic performance. A similar decrease in biogas yield with increasing oxygen loading has also been reported in previous studies [13]. This inhibitory effect may be attributed to the poor fluidity and mass-transfer characteristics of high-solids systems, as well as the destabilization of anaerobic microbial communities under excessive oxygen exposure. Overall, the effects of micro-aeration on methane yield were strongly dose dependent, with improvement observed primarily under low oxygen loading. However, excessive oxygen significantly inhibited reactor performance by reducing overall biogas production and methane yield. Furthermore, fermentation systems with higher-solid contents demonstrated lower tolerance to variations in oxygen loading.

3.2. The tendency of physicochemical properties during anaerobic digestion of food waste in micro-aerobic conditions

3.2.1. Variations of pH, ORP, NH_4^+-N, and SCOD during continuous digestion

Fig. 2 (A-D) illustrates the effects of varying oxygen loading rates on pH, ORP, NH_4^+-N, and SCOD during the anaerobic digestion of FW at different TS levels (6% and 10%).

ORP measurements showed changes in bulk redox conditions with oxygen input (Fig. 2B). At TS= 6%, ORP ranged from −374 to −346 mV during CK, from −368 to −351 mV at 2.625 mLO_2/L_R/d, and from −367 to −343 mV at 5.25 mLO_2/L_R/d. These overlapping ranges indicate that the lower oxygen inputs did not produce a clearly separated bulk redox regime. At 10.50 mLO_2/L_R/d, ORP increased to −328 mV by the end of the phase, while pH decreased to 7.07. At TS = 10%, ORP ranged from −378 to −369 mV during CK and from −373 to −341 mV at 2.625 mLO_2/L_R/d. Under the higher oxygen-loading conditions, greater fluctuations and an eventual increase toward approximately −332 mV were reported. The pH of the TS = 10% system remained within 7.32–7.68 over the experiment. Interpreted together with VFA profiles and methane production, the redox changes are consistent with a shift in the balance between upstream substrate conversion and downstream methane formation. The different responses may reflect the combined effects of solids content, OLR, and mass-transfer conditions. However, ORP integrates multiple redox processes and is influenced by pH; it does not quantify dissolved oxygen (DO) or establish spatial oxygen distributions. The treatments are therefore defined by nominal oxygen input, with ORP providing complementary evidence of the bulk redox response.

As depicted in Fig. 2(C), NH_4^+-N concentration remained at a relatively low level at TS = 6%. During the oxygenation phase of 2.625–5.25 mLO_2/L_R/d, the NH_4^+-N concentration showed minor fluctuations and remained relatively stable. This indicated that micro-aerobic conditions moderately promoted the hydrolysis and ammonification of nitrogen-containing macromolecules, and the release rate achieved a dynamic balance with microbial assimilation and methanogenic consumption. In the high-oxygen stage, the NH_4^+-N concentration exhibited substantial oscillations with a declining trend. This could be attributed to the elevated oxygen levels stimulating the activity of aerobic or facultative microorganisms, which accelerated the consumption of certain nitrogen sources or induced localized partial nitrification.

In TS= 10% system, the overall NH_4^+-N level increased significantly due to the higher substrate concentration. During the micro-aerobic stage, the introduction of trace oxygen accelerated the hydrolysis and release of nitrogenous organic matter, resulting in a steady upward trend in NH_4^+-N concentration. Under the medium-oxygen condition, mass transfer was limited by the high solid content, causing intense localized hydrolysis and acidogenesis. Consequently, the NH_4^+-N release rate notably exceeded the microbial assimilation rate, leading to a distinct NH_4^+-N peak in the system. During the 10.5 mLO_2/L_R/d stage, the NH_4^+-N concentration initially experienced a sharp decrease followed by a substantial increase. The temporary decline was likely due to the depletion of readily degradable nitrogenous substrates. The subsequent increase could be attributed to the intense micro-aerobic conditions further disrupting the structure of residual refractory proteins, thereby triggering deep hydrolysis, coupled with the cell death and autolysis of inhibited microorganisms. Furthermore, the variation pattern of NH_4^+-N concentration was consistent with the accumulation trend of SCOD, which further verified the potential risk of methanogenic inhibition in high-solid systems during the high-oxygen stage.

As shown in Fig. 2(D), in TS= 6% system, oxygen loads of 2.625–5.25 mLO_2/L_R/d accelerated substrate hydrolysis, supplying ample small-molecule substrates for methanogens. The slight inhibition of methanogens under these conditions was offset by the increased substrate availability, leading to a slow SCOD accumulation. Conversely, under high oxygen conditions (TS=10%), severe methanogenic inhibition blocked Soluble Organic Matter (SOM) degradation, resulting in a significant and continuous rise in SCOD. In TS= 10% system, SCOD remained relatively stable during the micro-aeration phase, indicating a temporary metabolic balance between substrate hydrolysis and methanogenesis. This led to massive accumulation of SOM, a sharp increase in SCOD, and a decline in methanogenic efficiency. At 10.5 mLO_2/L_R/d, the observed decrease in SCOD was mainly attributed to localized oxidative decomposition of soluble organics rather than methanogenic degradation [22]. These variations in SCOD are consistent with the findings regarding methanogenic efficiency discussed above. This may reflect an imbalance between substrate solubilization and downstream consumption under the higher oxygen-input condition.

3.2.2. Characteristics of VFA production and lignocellulose degradation during continuous digestion

Fig. 2 (E-H) demonstrates that the regulation of solids content played a significant role, with the efficacy of micro-aeration exhibiting a non-linear relationship with oxygen concentration, consistent with previous studies [9,23,24]. Total VFA (TVFA) concentrations in both R1 and R2 initially decreased before increasing.

In R1, TVFA levels dropped to a maximum of 1531.96 mg/L and an average of 1307.29 mg/L, remaining stable with no accumulation. However, high oxygen levels led to TVFA accumulation and the inhibition of methanogens, resulting in a decreased methane yield. In R2, TVFA decreased under the micro-aeration treatment, indicating that metabolic equilibrium was achieved [25]. Although TVFA remained stable at moderate oxygen levels, the synergistic inhibition of methanogens by high oxygen and high NH_4^+-N, combined with the effects of reactive oxygen species (ROS), reduced methanogenic efficiency. Acetic acid remained the predominant component of VFA in both systems, a characteristic unaltered by micro-aeration [26]. Under low and moderate oxygen conditions, acetic acid was consumed efficiently, and propionic acid was maintained at extremely low concentrations, suggesting that oxygen facilitated the conversion of propionate to methane [27].

At TS = 10%, degradation improved during phases R2_2 and R2_3; however, under the highest oxygen loading (10.50 mLO_2/L_R/d), the apparent removal values of cellulose, hemicellulose, and lignin became negative. These negative values indicate an apparent enrichment of the corresponding lignocellulosic fractions in the residual solids relative to the anaerobic baseline, rather than net formation of these components. Similar dose-dependent responses have been reported in lignocellulose-based anaerobic digestion, where moderate micro-aeration enhanced cellulose and hemicellulose degradation, whereas further increases in oxygen loading reduced methane production and disrupted the microbial community structure [14]. Moderate micro-aeration can stimulate lignocellulose-degrading microorganisms and enzymes, thereby improving the hydrolysis of cellulose and hemicellulose; however, excessive oxygen exposure may impose oxidative stress on obligate anaerobes, particularly methanogens, and consequently disturb the

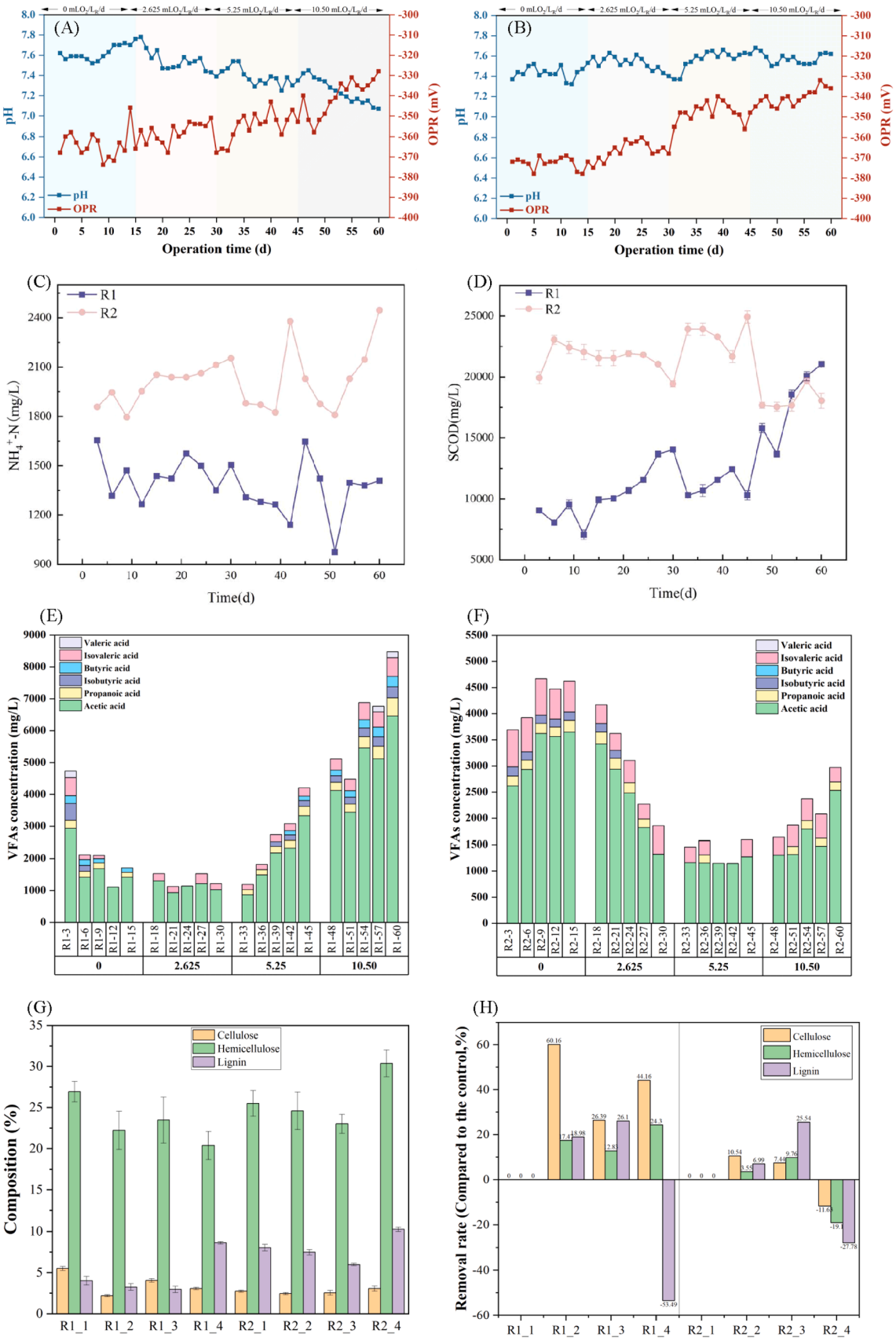


Fig. 2. Dynamics of physicochemical parameters, lignocellulose degradation, and VFA production during anaerobic digestion at different TS levels and oxygen loading rates.

balance between hydrolysis and downstream methanogenesis [13,18]. Under such conditions, the further conversion of recalcitrant lignocellulosic fractions may be limited, resulting in their increased relative proportion in the residual solids [28]. The effect was more pronounced in the TS= 10% system, suggesting a narrower oxygen-tolerance range under high-solids conditions, consistent with the gas production results in Section 3.1.

3.2.3. Analysis of hydrolysis, acidification, acetogenesis, and methanogenesis efficiencies during continuous digestion

Fig. 3. reveals distinct response patterns of process efficiencies to oxygen loads in the two systems, indicating that the regulatory effect of micro-aeration was closely correlated with solids content.

In the R1 system, hydrolysis efficiency under micro-aeration improved by 19.49%, 17.31%, and 28.15% compared to the anaerobic phase, which is consistent with the SCOD generation profile. Acidification efficiency initially decreased and then increased with rising oxygen loads, mirroring the trend observed in TVFA concentrations. Acetogenesis efficiency improved by 11.62% only at 5.25 $mLO_2/L_R/d$, aligning with the increased acetate content during this phase. The calculated methanogenesis efficiency increased at 2.625 and 5.25 $mLO_2/L_R/d$; however, the response of methane content differed between these two conditions. At 5.25 $mLO_2/L_R/d$, the increase in total biogas production was accompanied by a decrease in methane content, indicating that the enhanced overall gas production was primarily driven by intensified upstream substrate conversion rather than a proportional enhancement of methanogenesis. At 10.50 $mLO_2/L_R/d$, methanogenesis efficiency declined by 27.50%, consistent with the deterioration in methane production.

The efficiency variations in R2 differed from those in R1. Improvements in hydrolysis, acidification, acetogenesis, and methanogenesis efficiencies were achieved simultaneously only at 2.625 $mLO_2/L_R/d$, exceeding the anaerobic phase by 6.22%, 6.13%, 9.81%, and 16.23%, respectively. Conversely, at 10.50 $mLO_2/L_R/d$, all efficiencies dropped significantly by 24.69%, 36.53%, 35.50%, and 33.21% compared to the anaerobic phase, suggesting that the digestion stability of high-solids systems is more susceptible to the adverse effects of oxygen.

Integrating the efficiencies of the four conversion steps, it is recommended to control the micro-aeration dosage at 2.625–5.25 $mLO_2/L_R/d$ for low-solids FW AD. For high-solids systems, the dosage should not exceed 2.625 $mLO_2/L_R/d$. This conclusion aligns with the observed patterns of methanogenic efficiency.

The increase in methane production under appropriate micro-aeration may be attributed to the improved coordination between hydrolysis and methanogenesis [18]. Limited oxygen input promoted substrate solubilization and hydrolysis while avoiding substantial VFA accumulation, thereby increasing the availability of soluble intermediates for subsequent methanogenesis [13]. This interpretation is supported by the simultaneous improvement in hydrolysis and methanogenesis efficiencies under the appropriate oxygen-loading conditions.

3.3. Microbial community analysis

3.3.1. Microbial community composition at phylum and genus levels during continuous digestion

Fig. 4 (B-E) illustrates the characteristics of the microbial community at the phylum and genus levels in continuous anaerobic digestion of FW enhanced by micro-aeration.

At the bacterial phylum level, *Firmicutes*, *Bacteroidota*, *Actinobacteriota*, and *Synergistota* were the predominant phyla in both the R1 and R2 systems, accounting for a total relative abundance of 85.53%-96.85%. Their functional complementarity supported the digestion process. In the R1 system, the abundance of Firmicutes, the core acidogens, showed a decreasing trend with increasing oxygen concentration. However, the uniform mass transfer inherent to the low-solids environment ensured a basal abundance sufficient to maintain hydrolysis and acidification functions. The enrichment trend of *Bacteroidota* (which enhances substrate hydrolysis) aligned with the variations in SCOD generation and hydrolysis efficiency. *Actinobacteriota* has been reported to participate in lignocellulose degradation [29], while *Synergistota* is known to mediate the conversion of propionate to acetate. Although the total bacterial abundance was high during the high-oxygen phase, facultative anaerobes became enriched, and their aerobic metabolism competed for metabolic substrates with methanogens. This disrupted the metabolic balance, resulting in enhanced hydrolysis and acidification but inhibited methanogenesis. In the R2 system, changes in community composition were highly consistent with methanogenic efficiency trends. In phases R2_1 and R2_3, the abundance of Firmicutes was similar, and functional groups like *Bacteroidota* showed no significant enrichment, corresponding to unchanged methanogenic efficiency. Only in R2_2 did Firmicutes remain stable while *Bacteroidota* showed slight

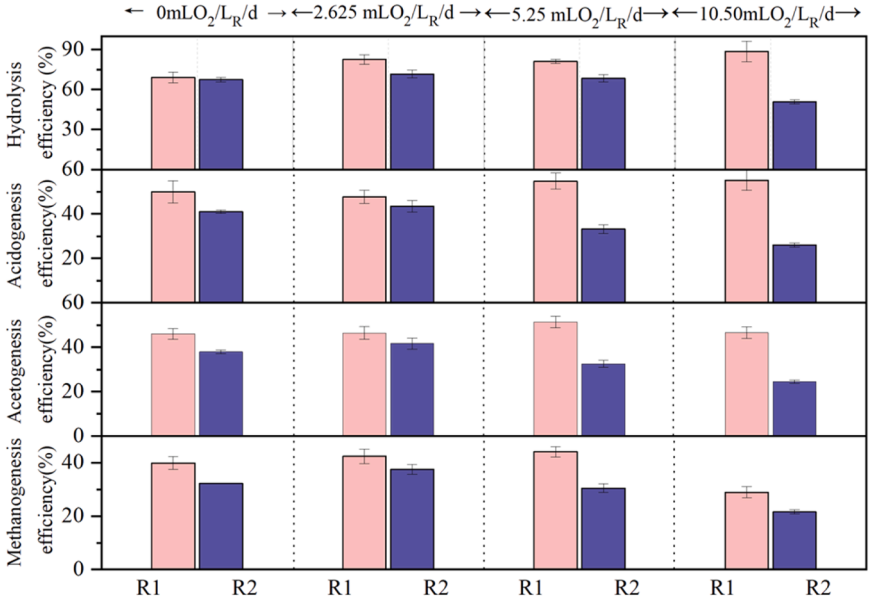


Fig. 3. Efficiency characteristics of the four anaerobic digestion stages at TS levels of 6% and 10% under different oxygen loading rates.

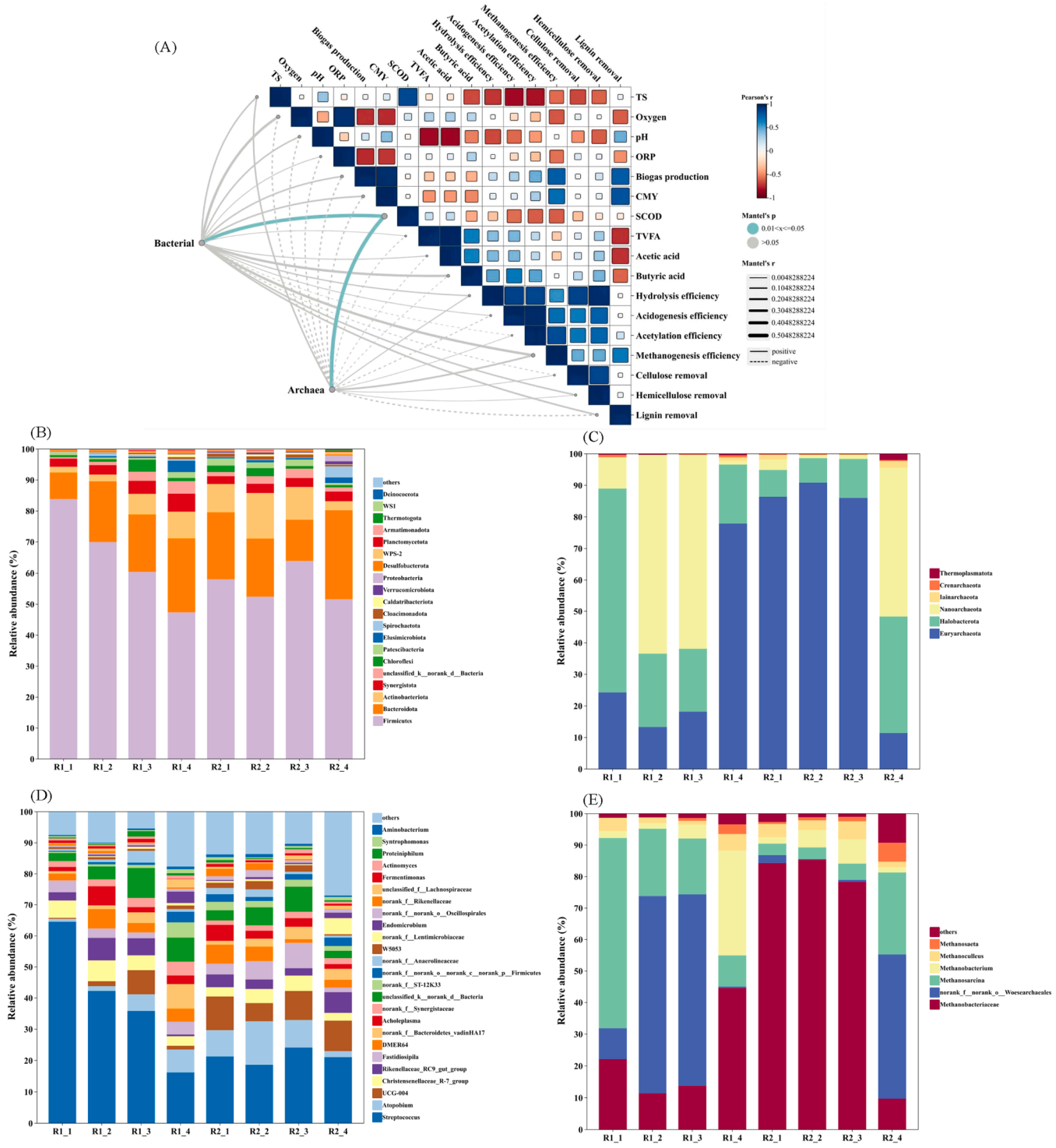


Fig. 4. Microbial community composition at the phylum and genus levels and its correlation with environmental factors based on Mantel test during continuous anaerobic digestion.

enrichment, forming a temporary equilibrium that significantly boosted methanogenic efficiency. Conversely, high oxygen levels led to an over-enrichment of *Bacteroidota* and a structural imbalance in the community, inhibiting methanogenesis. These results indicate that the beneficial effect of micro-aeration under the tested high-solids condition was mainly observed at the lowest oxygen loading.

At the archaeal phylum level, the response to oxygen differed between the two TS systems. Under TS= 6%, low and medium oxygen input enriched Nanoarchaeota, which may have supported Euryarchaeota by promoting organic matter degradation and supplying methanogenic substrates. However, high oxygen loading inhibited Halobacterota and Nanoarchaeota, while oxygen exposure may have interfered with methyl-coenzyme M reductase activity and intensified substrate competition, thereby reducing methane production. Under TS= 10%, aggregates and anoxic microenvironments favored the dominance of Euryarchaeota, and low oxygen loading could be rapidly consumed by facultative anaerobes, helping maintain methanogenic activity. In contrast, medium and high oxygen loading likely penetrated

the floc structure and created localized hyperoxic zones, suppressing methanogenic enzymes and favoring non-methanogenic archaea, ultimately decreasing methane production [30–35].

At the genus level, community shifts were consistent with the phylum-level patterns. Streptococcus, a dominant facultative genus within Firmicutes, remained stable under low oxygen but declined under high oxygen in the low-solids system, potentially weakening carbohydrate degradation. Micro-aeration enriched UCG-004 and Atopobium, supporting hydrolysis and VFA conversion. For archaea, Nanoarchaeota-related taxa were enriched at low-to-medium oxygen under TS= 6%, whereas Methanobacteriaceae/Methanobacterium in TS= 10% responded positively only to low oxygen, mainly via hydrogenotrophic methanogenesis. Medium and high oxygen inhibited methanogenic genera, confirming the broader oxygen tolerance of the low-solids system.

3.3.2. Correlation analysis

Mantel test analysis revealed a significant positive correlation between TS and SCOD (Fig. 4A). This is attributed to the increased degradation and digestion of organic matter available in high-solids systems. Conversely, TS showed a significant negative correlation with acidification and acetogenesis efficiencies. This was likely due to the lower mass transfer efficiency inherent in high-solids AD, which hindered substrate diffusion and inhibited the growth activity of acidogenic and acetogenic bacteria.

Oxygen exhibited a positive correlation with SCOD, TVFA, acetate, and cellulose removal, indicating that oxygen introduction facilitated organic matter degradation and TVFA generation. A significant negative correlation was observed between pH and both TVFA and acetate, confirming that the decline in system pH was driven by the substantial accumulation of VFA, particularly acetate. Biogas yield and cumulative methane yield (CMY) were significantly positively correlated with methanogenic efficiency and lignin removal rates, highlighting the critical role of lignocellulose degradation in methane production. Furthermore, significant positive correlations were found among the efficiencies of hydrolysis, acidification, acetogenesis, and methanogenesis, as well as with cellulose and hemicellulose removal rates, suggesting the four digestion steps were tightly interlinked.

Regarding the relationships between microorganisms and environmental factors, both bacteria and archaea showed significant positive correlations with SCOD, indicating their joint participation in substrate degradation. Oxygen was positively correlated with bacteria but negatively correlated with archaea. It is hypothesized moderate oxygen levels enhanced substrate hydrolysis by stimulating bacterial activity, while the produced intermediate metabolites could be further utilized by methanogens under anaerobic microenvironments. Conversely, archaea were negatively correlated with TVFA and positively correlated with methanogenic efficiency, reflecting their role in consuming organic acids to enhance methane production. Furthermore, bacterial communities were closely associated with hemicellulose and lignin removal, while archaeal communities showed stronger correlations with cellulose and hemicellulose degradation.

3.4. Prediction of functional genes and metabolic pathways

To further explore the potential functional basis of the observed responses to micro-aeration, PICRUSt2-based functional prediction and KEGG pathway analysis were conducted. These analyses provide predicted functional potentials rather than direct measurements of enzyme activities or metabolic processes. The potential effects of micro-aeration



Fig. 5. Predicted abundance of key enzyme-related genes and KEGG level 2 and level 3 metabolic pathways during micro-aeration-enhanced anaerobic digestion of food waste.

were therefore interpreted based on the predicted functional profiles together with the experimentally measured process performance (Fig. 5). First, the predicted abundance of lignocellulose-degrading enzymes, including β-glucosidase (EC 3.2.1.21) and β-mannosidase (EC 3.2.1.25), increased, which was consistent with the observed changes in cellulose and hemicellulose degradation. Second, the predicted abundance of genes encoding, enzymes related to electron transfer and methanogenesis, such as NADH:ubiquinone reductase and coenzyme F420 hydrogenase varied among the oxygen-loading treatments, suggesting potential changes in electron-transfer and methanogenic functions, suggesting potential differences in electron-transfer-related and methanogenesis-related functional capacity. The TS= 6% system was mainly associated with acetoclastic methanogenesis, whereas hydrogenotrophic methanogenesis was more prominent under TS= 10% [36–38]. Third, the predicted relative abundance of genes associated with antioxidant enzymes, including superoxide peroxidase (EC 1.11.1.15), superoxide dismutase (EC 1.15.1.1), and superoxide reductase (EC 1.15.1.2), increased under some micro-aeration conditions, suggesting potential changes in oxidative-stress-response capacity [39].

KEGG pathway prediction indicated that carbohydrate metabolism, amino acid metabolism, and energy metabolism were the dominant predicted functional categories. Changes in carbohydrate-metabolism-related gene profiles were broadly consistent with the measured lignocellulose conversion and hydrolysis responses. The predicted relative abundance of genes assigned to tricarboxylic acid (TCA)-cycle-related pathways increased in several micro-aeration groups, particularly in R1, suggesting differences in central metabolic potential [40]. Meanwhile, an increase in the predicted relative abundance of oxidative-phosphorylation-related genes was observed. First, the higher predicted abundance of oxidative-phosphorylation-related genes in R1 suggested potential differences in energy-metabolism capacity [40]. Compared with R2, the R1 system showed stronger metabolic adaptability to oxygen fluctuation, as reflected by the coordinated regulation of hydrolysis-related pathways, energy metabolism, and antioxidant responses. In R2, functional enhancement was mainly observed under low oxygen loading, whereas medium and high oxygen levels disrupted metabolic coordination and failed to generate effective methanogenic precursors. In addition, pathways related to cofactors, vitamins, and two-component systems may have supported CoA-mediated intermediate metabolism and microbial responses to environmental changes [41–44]. The predicted functional profiles were further compared with the independently measured process parameters, including SCOD, VFA dynamics, lignocellulose removal, ORP, and methane production. The consistency between these datasets was used to support the proposed functional interpretation, while the PICRUSt2 results were not considered direct validation of metabolic activity. Direct enzyme-activity measurements were available from the preliminary batch experiments, where micro-aeration increased β-glucosidase and lignin peroxidase activities. However, direct measurements of electron-transfer activity and TCA-cycle activity were not conducted in the continuous reactors; therefore, the corresponding PICRUSt2 results are interpreted as potential functional responses rather than direct evidence of increased

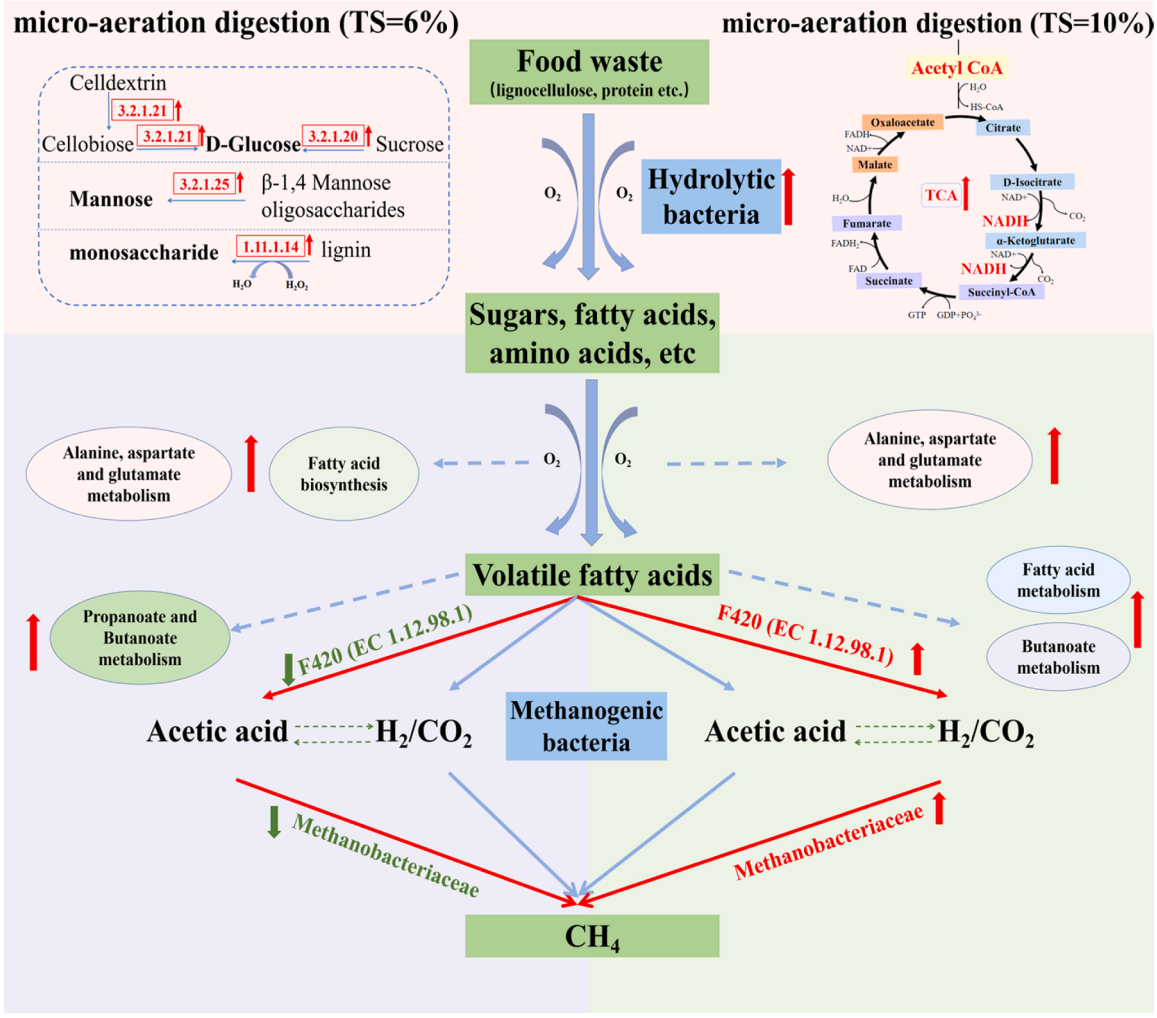


Fig. 6. Potential mechanisms of methane production during micro-aeration enhanced anaerobic digestion of food waste.

metabolic activity.

3.5. Proposed interpretation for methane production in micro-aeration enhanced anaerobic digestion of food waste

Based on the integrated analyses of reactor performance, physicochemical characteristics, microbial succession, enzyme prediction, and KEGG pathways, the response to micro-aeration was governed by a balance between accelerated upstream substrate conversion and preservation of downstream methanogenic capacity (Fig. 6). Appropriate oxygen input altered the abundance of functional microorganisms such as Firmicutes and increased the predicted abundance of hydrolysis-related functions, including α-glucosidase (EC 3.2.1.20), thereby accelerating lignocellulose decomposition and soluble substrate release [45]. In the TS = 6% system, the observed response may be associated with more favorable mass-transfer conditions, which could allow moderate oxygen input to promote upstream substrate conversion without severe inhibition of methanogenesis. In contrast, higher TS= 10% system showed a narrower apparent favorable response to oxygen loading. This greater sensitivity may be associated with the higher solids content, altered rheological and mass-transfer characteristics, and greater spatial heterogeneity under the TS = 10% condition. Microbial community shifts and PICRUSt2-derived functional predictions were generally consistent with this process-level interpretation, which showed a higher predicted abundance of TCA-cycle-related functions and improved oxidative-stress tolerance. However, excessive oxygen disrupted this balance, leading to oxidative stress, VFA accumulation, and methanogenic inhibition. Overall, the beneficial effect of micro-aeration depended on matching oxygen dosage with solids content, enabling simultaneous enhancement of substrate degradation and methane production. It should be noted that the functional profiles obtained from PICRUSt2 represent predictions based on 16S rRNA gene data rather than direct measurements of gene expression, enzyme activity, or metabolic flux. Therefore, the predicted functional changes should be interpreted as potential metabolic associations. In this study, these predictions were interpreted together with the experimentally measured process performance, VFA profiles, lignocellulose degradation, and microbial community data to provide a mechanistic hypothesis rather than definitive evidence of pathway activation. Overall, the present results support a conceptual framework in which the benefit of micro-aeration depends on maintaining a balance between oxygen-stimulated upstream carbon conversion and oxygen-sensitive downstream methanogenesis. Direct validation using metagenomic/metatranscriptomic analyses and targeted enzyme or metabolite measurements would be valuable in future studies.

3.6. Preliminary energy assessment

A preliminary energy balance was assessed from the experimental data. The energy input was calculated from the operating voltage of the peristaltic pump (10 V) and the recorded average current (15 mA). For the calculated daily aeration durations of 100, 200, and 400 min, the corresponding pump electricity inputs are 0.25, 0.50, and 1.00 Wh/d, respectively. These estimates apply to both TS conditions and assume that the stated voltage and current represent the actual electrical input during aeration; they exclude standby, mixing, heating, and other auxiliary electricity. They should therefore be interpreted as conditional estimates rather than measured total reactor energy consumption. At TS = 6%, the net energy gains relative to the anaerobic control were approximately 2.34 and 1.99 Wh/d at oxygen loading rates of 2.625 and 5.25 $mLO_2/L_R/d$, respectively, whereas a net energy loss of 12.66 Wh/d was observed at 10.5 $mLO_2/L_R/d$. At TS = 10%, the lowest oxygen loading rate provided the greatest net energy gain of approximately 9.22 Wh/d, while higher oxygen loading rates resulted in negligible or negative energetic benefits.

Therefore, the present calculation should be regarded as a preliminary estimate of the additional electricity requirement for air delivery rather than a complete energy or economic assessment. The results suggest that controlled micro-aeration may provide a favorable energy balance under the tested conditions, although pilot- and full-scale studies are still necessary to assess oxygen transfer, mixing, total energy consumption, and long-term stability with precisely controlled dosing [46].

4. Conclusion

This study demonstrated that the effectiveness of micro-aeration on continuous FW AD was strongly dependent on oxygen loading and TS content. Low oxygen loading of 2.625 $mLO_2/L_R/d$ resulted in a modest improvement in methane production and process stability in both TS systems, while the TS = 6% system tolerated a wider oxygen range of 2.625–5.25 $mLO_2/L_R/d$. Under appropriate micro-aeration, lignocellulose degradation was promoted, with cellulose removal reaching 60.16% and lignin removal increasing by up to 26.10%. In contrast, excessive aeration at 10.50 $mLO_2/L_R/d$ adversely affected process stability, as indicated by increased ORP, VFA accumulation, pH decline, SCOD fluctuation, reduced methane production, and weakened lignocellulose degradation. The TS= 10% system showed lower oxygen tolerance, likely due to restricted mass transfer and greater spatial heterogeneity. Microbial community and predicted functional analyses suggested that appropriate oxygen input favored hydrolysis-related bacteria, methanogens, and increased the predicted potential for hydrolysis, electron transfer, and oxidative-stress response. Overall, the findings highlight the importance of matching oxygen input with reactor loading conditions to maintain coordination between upstream substrate conversion and downstream methane formation. Micro-aeration exerted distinct effects on hydrolysis and methanogenesis: low oxygen loading could modestly improve methane production while maintaining methanogenic stability, whereas moderate oxygen loading primarily promoted upstream substrate conversion without a proportional increase in methane formation.

CRediT authorship contribution statement

Ziyu Xiong: Writing – review & editing, Writing – original draft, Visualization, Resources, Formal analysis, Data curation. **Yutong Wu:** Writing – review & editing, Writing – original draft, Visualization. **Jinglin Li:** Software, Resources, Formal analysis. **Yafan Cai:** Supervision, Investigation, Conceptualization. **Xingyao Meng:** Writing – original draft, Resources, Funding acquisition, Data curation. **Pan Wang:** Validation, Supervision, Investigation. **Lianhai Ren:** Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] UNEP, Food Waste Index Report 2024: Think Eat Save, Tracking Progress to Hal ve Global Food Waste, UN, Nairobi, 2024.
[2] G. Capson-Tojo, M. Rouez, M. Crest, J.-P. Steyer, J.-P. Delgenès, R. Escudié, Food waste valorization via anaerobic processes: a review, Rev. Environ. Sci. Biotechnol. 15 (2016) 499–547.
[3] Y. Ren, M. Yu, C. Wu, Q. Wang, M. Gao, Q. Huang, Y. Liu, A comprehensive review on food waste anaerobic digestion: research upd ates and tendencies, Bioresour. Technol. 247 (2018) 1069–1076.
[4] L. Yang, L. Chen, C. Zhao, H. Li, J. Cai, Z. Deng, M. Liu, Biogas slurry recirculation regulates food waste fermentation: effects and mechanisms, J. Environ. Manag. 347 (2023) 119101.
[5] Z. Zhao, Y. Zhang, Y. Li, X. Quan, Z. Zhao, Comparing the mechanisms of ZVI and Fe3O4 for promoting waste-activate d sludge digestion, Water Res. 144 (2018) 126–133.
[6] L. Zhu, B. Wu, Y. Liu, J. Zhang, R. Deng, L. Gu, Strategy to enhance semi-continuous anaerobic digestion of food waste by combined use of calcium peroxide and magnetite, Water Res. 221 (2022) 118801.
[7] S.-F. Fu, K.-Q. Chen, W.-X. Sun, R. Zhu, Y. Zheng, H. Zou, Improved methane production of corn straw by the stimulation of calciu m peroxide, Energy Convers. Manag. 164 (2018) 36–41.
[8] S.-F. Fu, K.-Q. Chen, R. Zhu, W.-X. Sun, H. Zou, R.-B. Guo, Improved anaerobic digestion performance of Miscanthus floridulus by d ifferent pretreatment methods and preliminary economic analysis, Energy Convers. Manag. 159 (2018) 121–128.
[9] R. Zhu, D.-H. Wang, Y. Zheng, Understanding the mechanisms behind micro-aeration to enhance anaerobi c digestion of corn straw, Fuel 318 (2022).
[10] F.-R. Amin, H. Khalid, W.-W. Li, Enhanced methane production and energy potential from rice straw by em ploying microaerobic pretreatment via anaerobic digestion, J. Clean. Prod. 296 (2021).
[11] R. Zhu, Y.-F. Zhang, H. Zou, The effects of micro-aeration on semi-continued anaerobic digestion of corn straw with increasing organic loading rates, Renew. Energy 195 (2022) 1194–1201.
[12] U. Ullah, A.S. Alkorbid, M. Jalalah, F.A. Harraz, A.I. Alalawy, S.H.A. Hassan, E.-S. Salama, Lab to large-scale applications of microaeration in anaerobic digestion: biowaste digestibility, microbiome nexus, and reactor stability, J. Environ. Chem. Eng. 13 (2025) 117766, https://doi.org/10.1016/j.jece.2025.117766.
[13] X. Li, Y.J. Yan, C.S. Lu, H. Jiang, H. Ma, Y. Hu, Micro-aeration based anaerobic digestion for food waste treatment: a review, J. Water Process. Eng. 58 (2024) 104814.
[14] M.P. Kozera, Ç. Akyol, F. Bouchon, R. Ganigué, B. Willems, E. Meers, H.B. Møller, Dose-response effects of microaeration on thermophilic lignocellulose- based anaerobic digestion: performance, stability and microbial commun ity dynamics, Bioresour. Technol. Rep. (2026) 102593.
[15] Z. Jin, Q. Yu, Z. Zhao, Y. Zhang, Insight into the role of intermittent microaeration in enhancing lignocellulose-containing anaerobic digestion and humic substances formation, Chem. Eng. J. 520 (2025) 166500, https://doi.org/10.1016/j.cej.2025.166500.
[16] J. Zheng, X. Wang, Y. Zhang, Synergistic promotion of straw anaerobic digestion by a composite microbial consortium and micro-aeration: enhancing lignocellulose decomposition and methanogenesis, Chem. Eng. J. 528 (2026) 172265, https://doi.org/10.1016/j.cej.2025.172265.
[17] N. Yu, Different micro-aeration rates facilitate production of different end-products from source-diverted blackwater, Water Res. 177 (2020) 115783.
[18] S.F. Fu, F. Wang, X.Z. Yuan, Micro-aeration: an attractive strategy to facilitate anaerobic digesti on process, Trends Biotechnol. 41 (2023) 1332–1344.
[19] L.V. Febroni, A.P. Rosa, A.C. Borges, Microaeration enhances methane production in anaerobic digestion: a systematic review and meta-analysis, Bioresour. Technol. 459 (2026) 135162, https://doi.org/10.1016/j.biortech.2026.135162.
[20] W. Chuenchart, C. Sawaya, K.C. Surendra, A.L. Smith, S.K. Khanal, In-situ intermittent micro-aeration in food waste and sewage sludge anaerobic co-digestion: performance, stability, and microbial dynamics, Bioresour. Technol. 427 (2025) 132398, https://doi.org/10.1016/j.biortech.2025.132398.
[21] S. Zhang, W. Chen, Y. Liu, P. Luo, H. Gu, A modified method for the accurate determination of chemical oxygen demand (COD) in high chloride oilfield wastewater, Open. J. Yangtze Oil Gas. 03 (2018) 263–277, https://doi.org/10.4236/ojogas.2018.34023.
[22] Z. Zhou, Q. Ming, Y. An, Performance and microbial community analysis of anaerobic sludge diges tion enhanced by in-situ micro aeration, J. Water Process. Eng. 42 (2021).
[23] C. Sawatdeenarunat, S. Sung, S.-K. Khanal, Enhanced volatile fatty acids production during anaerobic digestion of lignocellulosic biomass via micro-oxygenation, Bioresour. Technol. 237 (2017) 139–145.
[24] J. Sun, T. Zhou, F. Yin, S. Wang, Anaerobic co-biodegradation of polyhydroxyalkanoate and swine manure f or volatile fatty acid production: the impact of C/N ratios and microb ial dynamics, Bioresour. Technol. 418 (2025) 131995.
[25] X.-Z. Wang, T. Yuan, Z.-F. Lei, Supplementation of O2-containing gas nanobubble water to enhance metha ne production from anaerobic digestion of cellulose, Chem. Eng. J. 398 (2020).
[26] Q. Cao, W. Zhang, T. Lian, Revealing mechanism of micro-aeration for enhancing volatile fatty aci ds production from swine manure, Bioresour. Technol. 365 (2022) 128140.
[27] Q. Cao, W. Zhang, T. Lian, Roles of micro-aeration on enhancing volatile fatty acids and lactic a cid production from agricultural wastes, Bioresour. Technol. 347 (2022) 126656.
[28] Y. Gong, Z. Jin, X. Wang, Y. Zhang, Improving methane production and 4-chlor-ophenol removal in anaerobic d igestion of corn straw by adding Phanerochaete chrysosporium and bioch ar under microaerobic conditions, Water Res. 270 (2025) 122845.
[29] Q. Liu, Z. Xie, S. Tang, Q. Xie, X. He, D. Li, Synthetic microbial community enhances lignocellulose degradation duri ng composting by assembling fungal communities, Bioresour. Technol. 419 (2025) 132068.
[30] P.N. Evans, J.A. Boyd, A.O. Leu, B.J. Woodcroft, D.H. Parks, P. Hugenholtz, G. W. Tyson, An evolving view of methane metabolism in the Archaea, Nat. Rev. Microbiol. 17 (2019) 219–232.
[31] J.-K. Jarett, S. Nayfach, M. Podar, Single-cell genomics of co-sorted Nanoarchaeota suggests novel putativ e host associations and diversification of proteins involved in symbio sis, Microbiome 6 (2018).
[32] L.-K. Cheng, C.-U. Zhao, X.-Y. Yang, Application of a dissolved oxygen control strategy to increase the exp ression of Streptococcus suis glutamate dehydrogenase in Escherichia c oli, World J. Microbiol. Biotechnol. 37 (2021).
[33] S. Wilde, A. Dash, A. Johnson, Detoxification of reactive oxygen species by the hyaluronic acid capsu le of group A Streptococcus, Infect. Immun. (2023).
[34] B. Chen, K.-F. Yu, L. Fu, The diversity, community dynamics, and interactions of the microbiome in the world's deepest blue hole: insights into extreme environmental response patterns and tolerance of marine microorganisms, Microbiol. Spectr. 11 (2023) e00531-23.
[35] A. Tholen, M. Pester, A. Brune, Simultaneous methanogenesis and oxygen reduction by Methanobrevibacter cuticularis at low oxygen fluxes, FEMS Microbiol. Ecol. 62 (2007) 303–312.
[36] Y.-S. Shi, H.-H. Xue, Y.-L. Yao, Overcoming methanogenesis barrier to acid inhibition and enhancing PAH s removal by granular biochar during anaerobic digestion, Chem. Eng. J. 477 (2023).
[37] T.-Y. Zhang, J.-X. Zhang, P.-S. Zhang, Electronic bifurcation: a new perspective on fe bio-utilization in ana erobic digestion of lactate, Environ. Sci. Technol. 57 (2023) 10448–10457.
[38] W. Li, Y.-L. Liu, B.-C. Wu, Upgrade the high-load anaerobic digestion and relieve acid stress thro ugh the strategy of side-stream micro-aeration: biochemical performanc es, microbial response and intrinsic mechanisms, Water Res. 221 (2022).
[39] Z.-Y. Wu, D. Nguyen, T.Y.C. Lam, Synergistic association between cytochrome bd-encoded Proteiniphilum a nd reactive oxygen species (ROS)-scavenging methanogens in microaerobi c-anaerobic digestion of lignocellulosic biomass, Water Res. 190 (2021).
[40] A. MacLean, F. Legendre, V.-D. Appanna, The tricarboxylic acid (TCA) cycle: a malleable metabolic network to c ounter cellular stress, Crit. Rev. Biochem. Mol. Biol. 58 (2023) 81–97.
[41] W. Ye, L. Li, P. Yang, Z. Zhang, L. Han, R. Yuan, X. Peng, Mechanism of extracellular polymeric substance production and its cont ribution to anaerobic digestion foaming: insights from cellular signal ling, Chem. Eng. J. 500 (2024) 157377.
[42] N. Ismail, N. Kureishy, S.-J. Church, Vitamin B5 (d-pantothenic acid) localizes in myelinated structures of the rat brain: potential role for cerebral vitamin B5 stores in local myelin homeostasis, Biochem. Biophys. Res. Commun. 522 (2020) 220–225.
[43] T. Zhang, P. Zhang, Z. Hu, New insight on Fe-bioavailability: bio-uptake, utilization and induce in optimizing methane production in anaerobic digestion, Chem. Eng. J. 441 (2022) 136099.
[44] T. Yao, X. Liu, D. Li, Y. Huang, W. Yang, R. Liu, Y. Pang, Two-component system RstAB promotes the pathogenicity of adherent-inva sive Escherichia coli in response to acidic conditions within macropha ges, Gut Microbes 16 (2024) 2356642.
[45] C.A. Alba Fierro, M.Á. Escobedo Bretado, D.M. Núñez Ramírez, M.A. Martell Nevárez, F.J. Ríos Fránquez, Are biological pretreatments of lignocellulosic residues a real option for biofuels production? BioResources 19 (2024).
[46] L. Costa, M.S. Duarte, C.P. Magalhães, M.A. Pereira, A.J. Cavaleiro, Micro-aeration for improving anaerobic treatment and biogas production from organic pollutants, Appl. Microbiol. Biotechnol. 109 (2025) 131, https://doi.org/10.1007/s00253-025-13519-w.